

University of Alberta
Department of Mathematics & Statistical Sciences

STAT 437 Applied Statistical Methods
Fall 2026

Instructor:	Sharandeep Singh Pandher
Office:	UCOMM 4-181B
Email:	sharand1@ualberta.ca
Lectures Room & Time:	BS M-149, MWF 9:00-9:50 am
Course Web Page:	Canvas
Office Hours:	In-person: MR 10.00-11.45 (UCOMM 1-180)

COURSE CONTENT

Course Description:

Topics covered Generalized Linear Models (GLMs) for non-normal responses (including binary, categorical, and counts); Mixed Effect Models for repeated measures, longitudinal or multilevel data; Bayesian Analysis; Nonparametric Regression, survival analysis. Applications to real data will be emphasized, including case studies and real-data applications.

1. Learn how to use modern statistical methods to answer scientific questions.
2. Develop data analytic skills including familiarity with R software.

Course Prerequisites: STAT 252 or STAT 378 or equivalent (STAT 337). Although the emphasis is on application, a general knowledge of statistical theory including likelihood theory, hypothesis testing, and confidence intervals, is also expected to be successful in this course.

Tentative Syllabus, Assigned Readings & Learning Resources

1. An introduction to multiple linear regression (MLR).

Recommended book: “*Generalized Linear Models with Examples in R*” by Dunn & Smyth 1st edition (2018)

Recommended Reading: Chapters 2 to 3

2. Generalized Linear Models (GLMs):
(i) Binary, Categorical, Counts responses,
(ii) Model Evaluation.

Recommended book: “*Generalized Linear Models with Examples in R*” by Dunn & Smyth, 1st edition (2018)

Recommended Reading: Chapters 4 to 11

3. Possible Extensions of MLR & GLM:
(i) Linear Mixed Effects Models
(ii) Generalized Linear Mixed Models.

Recommended book: “*Mixed Effects Models and Extensions in Ecology with R*” by Zuur et al, 1st edition (2009)

Recommended Reading: Chapters 5, 9, 12 & 13.

4. Survival analysis:
 - (i) Basic Quantities and Models
 - (ii) Nonparametric Survival Curve Estimation
 - (iii) Cox Proportional Hazards Model

Recommended book: *Survival Analysis* by Klein & Moeschberger, 2nd edition (2003)
 Recommended Reading: Chapters 2

5. Nonparametric Regression.

Recommended book: *The Elements of Statistical Learning* by Hastie, Tibshirani & Friedman, 2nd edition (2009)
 Recommended Reading: Chapters 6 & 7

6. Simulation Study and Bootstrap Methods

Recommended book: "Bootstrap Methods With Applications in R" by Gerhard Dikta & Marsel Scheer 1st edition (2021)
 Recommended Reading: Chapters 6

7. Bayesian Generalized Linear Mixed Models and Random Effects will be discussed in detail (If time permits).

Recommended book: *Statistical Rethinking* by McElreath, 2nd edition (2019)
 Recommended Reading: Chapters 11 & 13

Additional Recommended book: *Extending the linear Model with R*, by J. Faraway. An E-version of this book is available at the U of A library.

Note 1: The lecture notes provide sufficient material for all components of the course, including assignments, in-class quizzes, the midterm project, and the final project. The recommended books are optional and are provided as additional resources. The books are intended primarily for identifying and selecting suitable datasets for the midterm and final projects.

Software:

R is free and can be downloaded from: [http:// www.r-project.org/](http://www.r-project.org/).
 A tutorial for R: <https://cran.r-project.org/doc/manuals/r-intro.pdf>
 A website with useful R tips: [http:// www.statmethods.net/index.html](http://www.statmethods.net/index.html)

GRADE EVALUATION

The course mark will be calculated based on the following breakdown

Course Component	Weight	Date
Assignments	20%	See below
In-Class Quizzes	18%	See below
Data Analysis Project (Midterm Project)	15%	Oct 30
Oral Presentation of Data Analysis Project	7%	Nov 2, Nov 4 & Nov 6
Final Project	40%	Dec 18

Note 2: Please submit your Midterm Project and Final Project via Canvas before **10:00 pm** on the due date.

Note 3: Each student will be assigned randomly to a group of two or three students for Midterm and Final Projects.

Note 4: Please submit your Oral Presentation Slides of Midterm Project via Canvas before **10:00 pm, October 30.**

Note 5: For students in STAT 437, the final letter grade will be determined from the course mark as follows: An overall course mark of 50% or more guarantees a passing grade of at least D. An overall course mark of 90% or more guarantees a grade of at least A.

Grades are unofficial until approved by the Department and/or Faculty offering the course.

Assignments:

There are five equal-weighted assignments, a total of 20%. Please submit your assignments via Canvas before **10:00 pm** on the due date listed below.

Assignment No.	Weight of Total Mark	Due Date
Assign 1	4%	Sep 18
Assign 2	4%	Oct 2
Assign 3	4%	Oct 16
Assign 4	4%	Nov 20
Assign 5	4%	Dec 4

You can discuss assignments with others but must write your homework independently. Duplicated homework or solution with no supporting work receives no credit. The assignments are strongly recommended to be done using R-markdown if you used R codes.

In-Class Quizzes: There will be three in-class quizzes, each weighted equally, for a total of 18%. The quizzes will be based on the material covered in the assignments. The quizzes will be administered during scheduled class times. Please make sure to attend the class to complete the quizzes.

In-Class Quizzes No.	Weight of Total Mark	Due Date
In-Class Quizzes 1	6%	Sep 25
In-Class Quizzes 2	6%	Oct 9
In-Class Quizzes 3	6%	Nov 27

Lecture syllabus: all dates and topics are subject to change, as necessitated by illness, closures, or other unforeseen circumstances

Week of	A textbook chapter/Topic	Items of note*
August 31 st	An introduction to multiple linear regression	First-class September 2 nd
September 7 th	Generalized Linear Models (GLMs)	
September 14 th	Generalized Linear Models (GLMs)	Assignment 1 on September 18 th
September 21 th	Possible Extensions of MLR	
September 28 th	Possible Extensions of GLM	Assignment 2 on October 2 th
October 5 th	Survival analysis	
October 12 th	Survival analysis	Assignment 3 on October 16 th
October 19 rd	Nonparametric Regression	

October 26 th	Nonparametric Regression	Data Analysis Project on October 30 th Oral Presentation Slides on October 30 th
November 2 th	Oral Presentation of Data Analysis Project	
November 9 th	Fall Term Reading Week	
November 16 th	Simulation Study and Bootstrap Methods	Assignment 4 on November 20 th
November 23 th	Simulation Study and Bootstrap Methods	
November 30 th	Bayesian Generalized Linear Mixed Models and Random Effects	Assignment 5 on December 4 st
December 7 th	Bayesian Generalized Linear Mixed Models and Random Effects	
December 14 th	Final Project	Final Project is on December 18 th

Note 6: There are no classes Sep. 7 & 30, Oct. 12, and Nov. 9 – 13, 2026. The Fall 2026 term officially ends on Tue, Dec. 8, 2026.

Past or Representative Evaluative Material:

Course materials, including lecture notes, R codes, data sets, assignments, projects, etc. will be posted on the course website.

General Comments:

Lecture notes, codes, and data sets will be posted on the course website. Assignments will be posted on the course website and announced in the class. Assignments and Data Analysis projects will be discussed in class.

Missed Term Work:

A student who cannot complete an in-class quiz, a team assignment, or midterm project due to incapacitating illness, severe domestic affliction, or other compelling reasons can apply for an excused absence. To apply for an excused absence, you must contact the instructor through official university e-mail **within two working days** of missing the assessment or as soon as possible. If an excused absence is granted, then the weight of the missed component will be transformed into the final project. An excused absence is a privilege and not a right. There is no guarantee that an absence will be excused.

Misrepresentation of facts to gain an excused absence is a serious breach of the Code of Student Behavior. In all cases, instructors may request adequate documentation to substantiate the reason for the absence at their discretion.

STUDENT RESPONSIBILITIES

Academic integrity:

“The University of Alberta is committed to the highest standards of academic integrity and honesty, as well as maintaining a learning environment that fosters the safety, security, and the inherent dignity of each member of the community, ensuring students conduct themselves accordingly. Students are expected to be familiar with these standards regarding academic honesty and to uphold the policies of the University in this respect. Students are particularly urged to familiarize themselves with the provisions of the Code of Students Behavior (online at www.governance.ualberta.ca) and avoid any behavior which potentially results in suspicions of cheating, plagiarism, misrepresentation of facts, and/or participation in an offense. Academic dishonesty is a serious offense and can result in suspension or expulsion from the University.”

All forms of academic dishonesty are unacceptable at the University. Any suspected offense will be reported to the Faculty of Science. Anyone who is found in violation of the Code of Student Behavior may receive a sanction. Typical sanctions include conduct probation, a mark reduction or a mark of 0 on an assessment, a grade reduction or a grade of F in a course, a remark on the transcript, and a recommendation for suspension or expulsion.

Students are expected to familiarize themselves with the Academic Integrity resources (covering the topics of cheating, collaboration, plagiarism, and substantial assistance) on the website of the office of the Dean of Students.

Cell Phones:

Cell Phones are to be turned off during lectures, labs, and seminars.

Recording and /or Distribution of Course Materials:

Audio or video recording, digital or otherwise, of lectures, labs, seminars, or any other teaching environment by students is allowed only with the prior written consent of the instructor or as a part of an approved accommodation plan. Student or Instructor content, digital or otherwise, created and/or used within the context of the course is to be used solely for personal study and is not to be used or distributed for any other purpose without prior written consent from the content author(s).

Students Eligible for Accessibility- Related Accommodations:

Eligible students have both rights and responsibilities with regard to accessibility-related accommodations. Consequently, scheduling exam accommodations in accordance with Accessibility Resources deadlines and procedures is essential. Please note adherence to procedures and deadlines is required for U of A to provide accommodations. Contact Accessibility Resources for further information.

Learning and Working Environment:

The Faculty of Science is committed to ensuring that all students, faculty and staff are able to work and study in an environment that is safe and free from discrimination, harassment, and violence of any kind. It does not tolerate behaviour that undermines that environment. This includes virtual environments and platforms.

If you are experiencing harassment, discrimination, fraud, theft or any other issue and would like to get confidential advice, please contact any of these campus services:

- [Office of Safe Disclosure & Human Rights](#): *A safe, neutral and confidential space to disclose concerns about how the University of Alberta policies, procedures or ethical standards are being applied. They provide strategic advice and referral on matters such as discrimination, harassment, duty to accommodate and wrong-doings. Disclosures can be made in person or online using the [Online Reporting Tool](#).*
- [University of Alberta Protective Services](#): *Peace officers dedicated to ensuring the safety and security of U of A campuses and community. Staff or students can contact UAPS to make a report if they feel unsafe, threatened, or targeted on campus or by another member of the university community.*
- [Office of the Student Ombuds](#): *A confidential and free service that strives to ensure that university processes related to students operate as fairly as possible. They offer information, advice, and support to students, faculty, and staff as they deal with academic, discipline, interpersonal, and financial issues related to student programs.*
- [Office of the Dean of Students](#): *They can assist students in navigating services to ensure they receive appropriate and timely resources. For students who are unsure of the support they may need, are concerned about how to access services on campus, or feel like they may need interim support while you wait to access a service, the Dean of Students office is here to help.*

Feeling Stressed, Anxious, or Upset?

It's normal for us to have different mental health experiences throughout the year. Know that there are people who want to help. You can reach out to your friends and access a variety of supports available on

and off campus at the [Need Help Now](#) webpage or by calling the 24-hour Distress Line: 780-482-4357 (HELP).

Land Acknowledgement:

The University of Alberta respectfully acknowledges that we are situated on Treaty 6 territory, traditional lands of First Nations and Métis people.

To learn more about the significance of this land acknowledgement, please read [this](#) useful article and associated links to more information.

Policy/Disclaimer: Policy about course outlines can be found in [Course Requirements, Evaluations Procedures and Grading of the University Calendar](#). Any typographical errors in this Course Outline are subject to change and will be announced in class.

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